

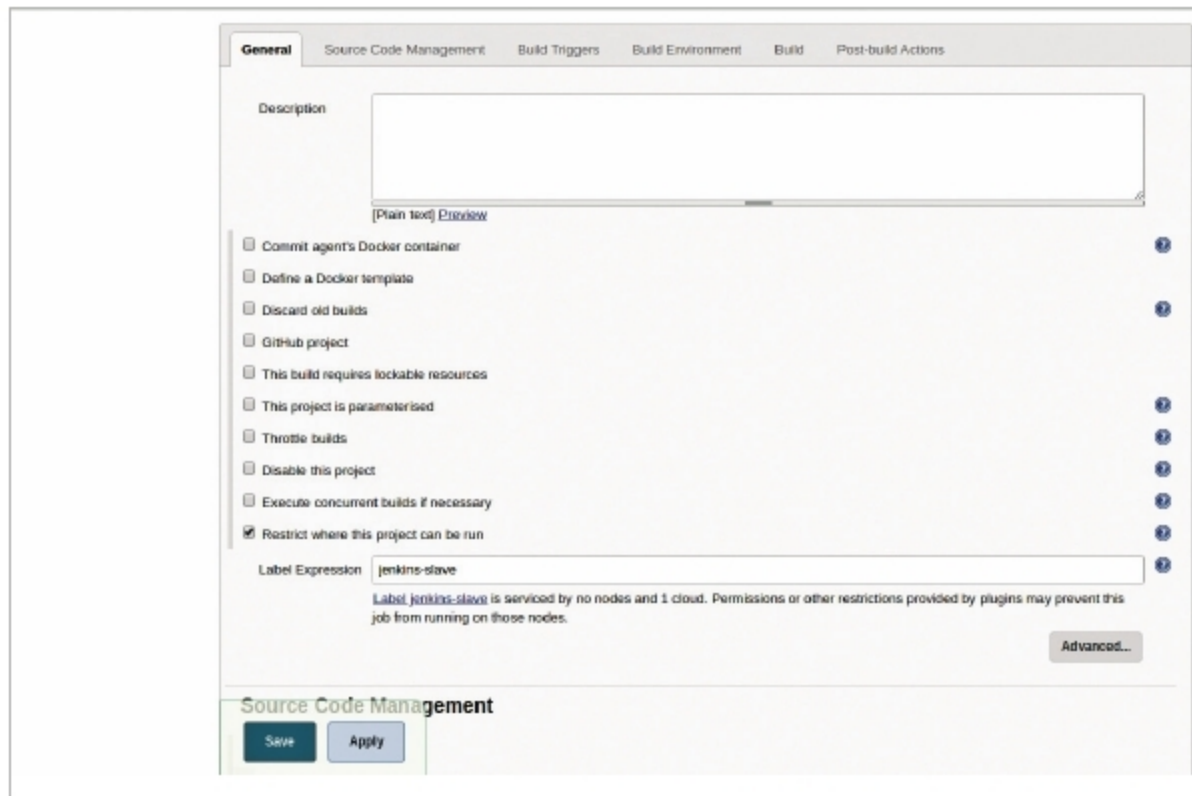


Figure 10: Configuring a job

container, it can be seen with the container details such as its name, id, created time, status, ports (Figure 5).

Configuring a Jenkins agent on Docker

Open a browser and go to <http://localhost:8080>. The Jenkins server UI will load. Once the initial configuration is done, its home page will look as shown in Figure 6.

Before we continue to set up the Jenkins Docker agent, we need to download the plugin for Docker. Go to *Manage Jenkins > Manage Plug-ins > Available*. Here, search in the filter for Docker; from the results, download the Docker plugin and click on *Install without Restart*. Once the plugin is successfully installed, go to *Manage Jenkins > Configure System*. Scroll down and look for the cloud, as shown in Figure 7.

Add a new cloud and select Docker. Now configure the Docker cloud details

and the Docker agent templates. In the Docker cloud section, give a name of your choice. Here, we have given the name as docker-master. In the Docker Host URI section, give the path of *docker.sock* so as to connect the Jenkins server to the Docker daemon. Check the enabled box so that it is available for use. With this, the Docker cloud/server has been configured. This configuration is shown in Figure 8.

It's time to configure the Docker agent template. The name for each agent template has to be different. The label can be the same for multiple agent templates, in case a group of agents is for one job and any can be used as per availability. But the Docker image name should be the same as the one pulled or built in the Docker system. This is shown in Figure 9. *Jenkins-agent* is the name of the Docker image which is built using the Jenkins/agent image. This image is configured with tools as per the application's requirement. This requirement can differ from application to application.

In the *Remote File System Root* column, give the location inside the container where it has to store the Jenkins Home data.

This is the most basic configuration of the Jenkins agent using Docker. Finally, apply and save the configuration. Now add a new item, give any project name and select any type of project. We have selected the *Freestyle* project. While configuring the job, look for *Restrict*, where the project can be run. Here, enter the template's name, which we have configured as shown in Figure 10. Configure the job as per the application. **END** 🐧

References

- [1] Jenkins 2.x Continuous Integration Cookbook - Third Edition; <https://www.amazon.com/gp/product/B0751HZ96H>
- [2] Jenkins Essentials - Second Edition; <https://www.amazon.com/gp/product/B073PGCKHY>
- [3] DevOps Bootcamp: The fastest way to learn DevOps; <https://www.amazon.com/gp/product/B07259Y9M8>
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